UNIVERSITY LIFE ASSISTANT

Survey of Models and LoRA / QLoRA Direction

Phase 1 · AI / Fine-tuning Survey · Report based on seed data and Kaggle baseline

Candidate Models (Evaluated on Kaggle 1x Tesla T4 16GB, 4-bit):

- **Qwen3.5-4B** *(Primary Candidate)*: Highest baseline exact match (8/10), optimal balance of quality and resource usage (3.6 GB peak VRAM).
- **Qwen3.5-2B** *(Lightweight Baseline)*: Fastest (2.9 s/sample), minimal VRAM footprint (1.9 GB), solid 7/10 exact match.
- **Gemma 3 4B-it** *(Cross-family Control)*: Google's 4B architecture to benchmark across model families against Qwen.
- **Llama 3.1 8B Instruct** *(Ceiling Benchmark)*: Upper-bound 8B parameter model to test whether scaling up justifies increased memory (6.4 GB VRAM).

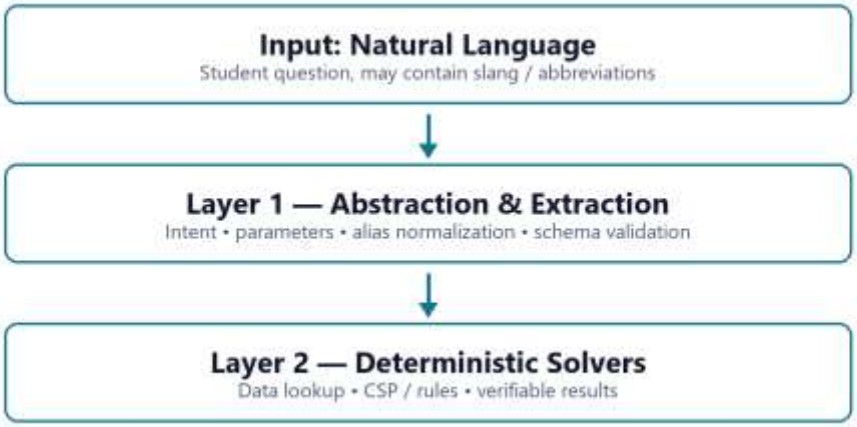
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1. Problem and Role in the System

Students need to make academic decisions while information may be scattered across syllabi, regulations, course schedules, and various other documents. Algorithmic solvers cannot directly utilize natural language queries containing slang, abbreviations, or missing information. This module handles the Abstraction step: transforming the query into a structured representation that can be validated and forwarded.

Role in the Two-Layer Architecture

Layer 1 receives the query, classifies the intent, extracts parameters, and normalizes aliases according to the catalog. Layer 2 uses the validated JSON to query data and execute deterministic logic such as rule-based checks or CSP/Knapsack where appropriate. The LLM must not autonomously infer prerequisites, schedule conflicts, or regulations without verifying against data/logic.



Flow example: "Gonna take OS wed morn, any clash with my current timetable?"

→ intent `check_schedule_conflict` and course/day/period fields → Layer 2 cross-checks against the actual timetable.

Scope

Focuses on converting Natural Language → JSON/DSL for the four defined intents; evaluating the prompt-only baseline; investigating the suitability of small models and the feasibility of using LoRA/QLoRA on T4 GPUs. Extracting regulations from PDFs and solving full optimization problems belong to other modules/phases.

2. Task Design: Natural Language → JSON

The output has the form {intent, params}. The model records only the information explicitly expressed in the sentence; it does not fabricate parameters and does not replace the data retrieval step. C

Intents and Parameters

Intent	Parameters / Types	Meaning
`check_prerequisite`	`target_course: string|null` `prerequisite_courses: string[]`	Course to check; prerequisite courses mentioned in the sentence.

`check_schedule_conflict`	`courses: string[]`
	`section: string|null`
	`day: enum|null`
	`period: enum|null`
	`start/end: HH:MM|null`
		Course, course section, and time to check; day = mon…sun; period expected morning/afternoon (needs expansion/normalization).

Intent	Parameters / Types	Meaning

`advise_academic_regulation`	`topic: enum`
	`standing/year: enum|null`
	`major: string|null`
	`gpa: number|null`
	`credits: number|null`
		Topic: max_credits, graduation_requirements, min_gpa, retake_limit, thesis_registration, expulsion_rules, other; extract standing/year/major/GPA/credits if stated.

`unknown`	`params: {}`	Out-of-scope query or insufficient information to reliably map to a business intent.

Normalization Conventions

• **Closed Enum:** Enum fields only accept values from the finalized list; course aliases are mapped to canonical names in the catalog (e.g., DSA → Data Structures). Aliases not in the catalog must be flagged/sent for clarification rather than silently accepted.

• **null and []:** null is used for single fields that are not stated/unspecified; [] is an empty list when no elements are extracted. For prerequisite_courses, [] does not mean the course definitely has no prerequisites — that is the conclusion of the data lookup step.

• **Out-of-domain:** Use {"intent":"unknown","params":{}} for out-of-scope requests. For ambiguous queries, the schema currently lacks a needs_clarification field; a unified decision is needed on whether to retain unknown or add a clarification flag in schema v1.

• **No inference:** Do not arbitrarily fill in standing, prerequisite, or time if not provided in the sentence. Schema checking must validate keys, enums, course names, and data types (e.g., GPA must be a number, not a string).

Input/Output Examples

Standard Query	Colloquial / Abbreviated Query
Input: "Do I need to pass Introduction to Programming before taking Data Structures?" `{"intent":"check_prerequisite","params":{"target_course":"Data Structures","prerequisite_courses":["Introduction to Programming"]}}`	*Input: "what's the prereq for DSA?"* `{"intent":"check_prerequisite","params":{"target_course":"Data Structures","prerequisite_courses":[]}}`

3. Seed Data and Evaluation Protocol

Workbook CT_Intent_Seed_Data.xlsx contains 43 labeled samples and a Course Catalog comprising 15 course names. The dataset is an exploratory seed constructed by the team, not a benchmark representative of all student questions.

Distribution by Intent

Intent	Sample Count	Train	Test
`advise_academic_regulation`	11	9	2

Intent	Sample Count	Train	Test
check_prerequisite	10	7	3
check_schedule_conflict	12	9	3
unknown	10	8	2
Total	**43**	**33**	**10**

Distribution by Case Type

Type	Total	Train	Test
Edge Case	16	11	5
Happy Path	17	14	3
Out-of-Domain	8	7	1
Vague	2	1	1

Protocol Used in Baseline

- 33 train samples and 10 test samples. Few-shot prompt used S02, S11, S17, S28, S10; these examples reside within the train split and do not overlap with the test answers according to experimental records.
- Test consists of S01, S06, S12, S14, S23, S32, S34, S39, S41, S43; encompassing all four intents, abbreviated/colloquial sentences, and one out-of-domain instruction-style query.
- Case types: Happy Path = 17; Edge Case = 16; Out-of-Domain = 8; Vague = 2.

4. Candidate Models and Selection Criteria

Selection criteria include: (i) model size testable on Kaggle T4 with 4-bit quantization; (ii) availability of an instruction-tuned version; (iii) stable JSON generation in baseline; (iv) PEFT/LoRA capability; (v) clear license and access terms; (vi) diversity across model families.

Model	Size / family	License	Role & Rationale for Fit	Limitations
Qwen3.5-2B	2B Qwen	Apache-2.0	Lightweight baseline; 7/10 exact match baseline, 1.9 GB peak VRAM, and 2.9 s/sample; useful for cost/quality trade-off comparison.	1 exact match sample behind Qwen3.5-4B on small test set; fine-tuning not yet attempted.
Qwen3.5-4B	4B Qwen	Apache-2.0	Primary initial candidate; highest baseline at 8/10 exact match, 3.6 GB peak VRAM. Good balance between performance and resources in this experiment.	Only 1 sample ahead of 2B model; difference is not statistically significant; Qwen 2B/4B are not independent families.
Gemma 3 4B-it	4B Gemma	Gemma Terms / custom	Cross-family candidate at the 4B mark; 7/10 exact match baseline, 3.6 GB VRAM; enables comparison across model families.	9/10 Schema OK; Google Gemma terms of use need to be accepted and reviewed.
Llama 3.1 8B Instruct	8B Llama	Llama 3.1 Community License (custom)	Larger ceiling benchmark to observe whether increasing model size improves performance; 7/10 exact match.	6.4 GB VRAM without improving exact match; HF access may require approval; model card lists officially supported languages excluding Vietnamese.

5. Baseline Experiment

Objective: Measure Natural Language → JSON conversion capability via prompting prior to fine-tuning, verify feasibility on Kaggle T4, and identify remaining errors.

Setup

Item	Configuration Used
Environment	Kaggle Notebook, single NVIDIA Tesla T4 GPU; batch size 1.
Weight Precision	4-bit NF4 quantization via bitsandbytes; compute fp16.
Prompt	System prompt containing 4 intents, parameter schemas/enums, null/[] conventions, and Course Catalog; plus 5 few-shot examples.
Few-shot	S02, S11, S17, S28, S10; all belong to the train split.
Decoding	Greedy: do_sample=False; max_new_tokens=160.
Test	10 samples: S01, S06, S12, S14, S23, S32, S34, S39, S41, S43.
Libraries	Transformers 5.19.0 according to original report logs.

Baseline Results

Model	Valid JSON	Clean	Schema OK	Intent acc.	Exact match	s/sample	Peak VRAM
Qwen3.5-2B	10/10	10/10	10/10	10/10	**7/10 (70%)**	2.9	1.9 GB
Qwen3.5-4B	10/10	10/10	10/10	10/10	**8/10 (80%)**	4.2	3.6 GB
Gemma 3 4B-it	10/10	10/10	9/10	10/10	**7/10 (70%)**	4.2	3.6 GB
Llama 3.1 8B Instruct	10/10	10/10	9/10	10/10	**7/10 (70%)**	3.4	6.4 GB

Metric Definitions

Valid JSON: Parsed valid JSON object. Clean: Contains only JSON without external conversational text. Schema OK: Correct keys matching intent, valid enums, and course names present in catalog (current validator does not yet fully enforce data types). Intent accuracy: Predicted intent matches gold label. Exact match: Entire JSON object matches gold JSON under strict comparison. Time/sample and peak VRAM measured during generation.

Interpretation

Prompting proved sufficient to generate valid JSON and correctly route intents across all 10 test samples; remaining errors lie in parameters and conventions. Qwen3.5-4B achieved the highest exact match in this run (8/10), whereas Qwen3.5-2B scored one sample lower while consuming roughly half the VRAM. Llama 3.1 8B consumed 6.4 GB without yielding a higher exact match. With n=10, these serve solely as directional signals; a single sample represents 10 percentage points.

6.1 Error Analysis

Sample / Error	Observation in Baseline	Implications
S39: "GPA needed for honors"	All four models chose graduation_requirements; gold label is currently other.	Review intent boundary / gold label prior to fine-tuning; this sentence may align better with graduation_requirements depending on definition.
S23: GPA = 1.5, asking about risk of expulsion	Some models autonomously added standing=academic_warning; Gemma returned GPA as string "1.5".	Disallow inferring unmentioned fields and add type validation to enforce GPA as a number.
S06: alias DSA	Gemma and Llama retained "DSA" instead of "Data Structures" and added an unmentioned prerequisite course.	Reinforce alias normalization and no-hallucination rules; schema checker must enforce canonical names.
S34: time 6 PM–8:30 PM	Some models inferred period=evening even though exact time was provided; Llama returned end=19:30 instead of 20:30.	Clarify precedence of exact time; test time parsing and prohibit self-altering time values.

Technical Limitations

- Test set contains only 10 samples, including 2 unknown/vague queries; results are not statistically robust.
- Catalog contains only course names present in the seed data, making in-prompt normalization easier than real-world deployment.
- Results depend on a single prompt/few-shot set; prompt optimization was not conducted individually per model.
- Both Qwen models ran with fallback implementation for certain attention kernels according to original logs; measured speed may be suboptimal.
- Schema checker did not catch incorrect data types, so "Schema OK" does not equate to full schema validation.

References

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